

HOSTEL UTILITY MONITORING

Project Report&Dashboard Analysis

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1. Executive Summary

This report presents a comprehensive analysis of the Hostel Utility Monitoring dashboard developed using Microsoft Excel. The dashboard consolidates energy consumption data across multiple sub-metering points, tracking electrical parameters including active power, reactive power, global intensity, and voltage over a period spanning late 2006 through mid-2007. The system enables facility managers to identify usage patterns, detect anomalies, and make data-driven decisions to optimize energy consumption and reduce operating costs.

Key highlights from the dashboard analysis:

- Energy consumption trends were tracked across three distinct sub-metering zones (Sub_metering_1, Sub_metering_2, Sub_metering_3).
- Global active power peaked significantly in January 2007, indicating heightened winter energy demand.
- Voltage remained consistently within the 230–255V range, suggesting stable power quality throughout the monitoring period.
- Sub_metering_3 accounted for the largest share of energy consumption (43%) based on the donut chart distribution.
- Hourly voltage and intensity data revealed predictable intraday usage cycles useful for demand-side management.

2. Project Overview

2.1 Background

The Hostel Utility Monitoring project was initiated to address the need for granular, real-time visibility into electrical consumption across a residential hostel facility. With rising energy costs and sustainability goals, the management required a tool to monitor sub-circuit level usage and compare it against global power readings.

2.2 Objectives

- Monitor and visualize energy consumption across all sub-metering points in real time.
- Track global active power, reactive power, global intensity, and voltage on a continuous basis.
- Identify peak usage periods, seasonal trends, and anomalies.
- Provide management with actionable insights to optimize energy usage and reduce costs.
- Establish a baseline for future energy audits and sustainability reporting.

2.3 Scope&Data Coverage

Parameter	Details
Monitoring Period	Late 2006 – July 2007 (~8 months)
Data Granularity	Hourly, Daily, Monthly aggregations

Metering Points	3 Sub-meters + Global meter
Key Variables	Active Power (kW), Reactive Power (kVAR), Voltage (V), Intensity (A)

3. Dashboard Components&Analysis

The Excel dashboard is organized into six distinct visualization panels, each targeting a specific analytical dimension of hostel utility consumption. Below is a detailed description of each component.

3.1 Sub-Metering Monthly Consumption (Grouped Bar Chart)

This chart displays the monthly sum of energy consumed across the three sub-metering zones from late 2006 through mid-2007. Sub_metering_1, Sub_metering_2, and Sub_metering_3 are rendered in distinct colors (blue, red, and green respectively), allowing quick visual comparison. January 2007 shows the highest combined sub-metering values, correlating with increased winter heating and lighting loads. The chart enables the facility team to monitor which zone is driving consumption growth month over month.

3.2 Global Power&Intensity Over Time (Multi-Line Chart)

This time-series line chart overlays three global electrical parameters: Global Intensity (A), Sum of Voltage (V), and Sum of Global Active Power (kW). The X-axis spans the full monitoring period at hourly resolution. A notable characteristic is the near-constant voltage line sitting well above the active power line, confirming that supply voltage remained stable while active power fluctuated significantly. Intensity follows a pattern closely mirroring active power, as expected from Ohm's Law relationships. Sharp spikes in power and intensity during early morning and evening hours indicate occupancy-driven demand peaks.

3.3 Average Global Power&Reactive Power (Bar Chart — Monthly)

Displayed in the top-right panel, this chart presents monthly averages of Global Active Power and Global Reactive Power. Active power averages are notably higher in December 2006 and January 2007, aligning with cold weather and reduced daylight. Reactive power follows a similar trend but at significantly lower magnitudes, indicating the hostel's loads are predominantly resistive (lighting, heating) with moderate inductive loads (motors, HVAC). Power factor improvement opportunities may exist in spring and summer months.

3.4 Voltage&Sub-Metering Stacked Bar Chart (Middle-Left)

This stacked bar chart visualizes the relationship between average voltage readings and sub-metering contributions across months. The stacked format clearly shows Sub_metering_3 as the consistently largest portion of the stack, followed by Sub_metering_2 and Sub_metering_1. Average voltage remains the background reference band, providing context for how power draw

correlates with supply voltage variations. The chart confirms that voltage stays within safe operational bounds even during months of high sub-meter consumption.

3.5 Sub-Metering Proportional Distribution (Donut Chart)

The donut chart on the right panel provides a percentage breakdown of total energy consumption by sub-metering zone and possibly by tariff category (0–10 scale in legend). Sub_metering_3 dominates with 43% of total consumption, Sub_metering_2 accounts for 38%, and Sub_metering_1 contributes 16%, with marginal unlabeled categories making up the remainder. This visualization is critical for pinpointing where targeted energy efficiency interventions would yield the greatest impact.

3.6 Hourly Voltage&Global Intensity (Stacked Bar — Bottom Panel)

The bottom panel presents an intraday profile across 24 hours (00:00–23:00) stacking Average of Global Intensity (orange) against Average of Voltage (blue). Voltage remains relatively constant across all hours (230–250V), while intensity shows elevated readings during morning (06:00–09:00) and evening (17:00–22:00) periods, clearly marking the occupancy-driven demand peaks. This chart is particularly valuable for time-of-use tariff analysis and scheduling high-load activities during off-peak hours.

4. Key Findings&Insights

4.1 Summary Metrics

Metric	Observed Value / Trend	Implication
Peak Active Power Period	January 2007	Winter heating is primary driver
Dominant Sub-Meter	Sub_metering_3 (43%)	Zone 3 is highest priority for efficiency measures
Voltage Stability	230–255 V (consistent)	Grid supply quality is reliable
Peak Demand Hours	06:00–09:00&17:00–22:00	Align scheduling to avoid demand charges
Reactive Power Ratio	Low vs. Active Power	Power factor is generally acceptable

5. Sub-Metering Analysis

The three sub-metering circuits represent distinct consumption zones within the hostel facility. Based on the dashboard data, the following breakdown and interpretation applies:

Zone	Share (%)	Likely End-Use	Priority Action
Sub_metering_1	16%	Kitchen / Laundry appliances	Schedule laundry off-peak
Sub_metering_2	38%	HVAC / Common areas	Optimize thermostat scheduling
Sub_metering_3	43%	Lighting / General outlets	Retrofit LED; install occupancy sensors

6. Voltage&Power Quality

Voltage quality analysis from the dashboard confirms that the hostel operates within safe electrical standards:

- Average voltage remains between 230V and 255V across all monitored hours and months, well within the IEC standard nominal range of 230V ±10%.
- No significant voltage sag or swell events are visible in the monthly or hourly profiles, indicating a reliable grid connection.
- Global reactive power remains low relative to active power, suggesting power factor values above 0.85 for most periods.

- The correlation between Global Intensity and Global Active Power confirms no significant harmonic distortion or measurement inconsistencies in the dataset.

These findings suggest that the hostel's electrical infrastructure is in good health and does not require immediate power quality remediation. Periodic power factor measurements are recommended as the facility scales operations.

7. Recommendations

7.1 Immediate Actions (0–3 Months)

- Conduct an energy audit of Sub_metering_3 (43% share) to identify specific high-consumption appliances and lighting loads.
- Shift high-load activities (laundry, bulk cooking) outside peak demand windows (06:00–09:00 and 17:00–22:00) to reduce demand charges.
- Install smart plugs or timers on common-area appliances connected to Sub_metering_1 to eliminate standby loads.

7.2 Medium-Term Actions (3–6 Months)

- Retrofit all common-area and corridor lighting with LED technology — estimated 40–60% reduction in lighting energy.
- Deploy occupancy sensors in bathrooms, common rooms, and corridors to automatically control lighting.
- Implement a Building Management System (BMS) integration with the existing monitoring dashboard for automated alerts on abnormal consumption.

7.3 Long-Term Strategy (6–12 Months)

- Investigate rooftop solar PV installation to offset daytime base load, particularly during low-demand spring/summer months.
- Upgrade the Excel dashboard to a real-time IoT-connected platform (e.g., Power BI, Grafana) for live monitoring and automated reporting.
- Establish monthly energy benchmarks per occupant and set progressive reduction targets aligned with sustainability goals.
- Consider a power factor correction capacitor bank if reactive power increases as HVAC loads expand.

8. Conclusion

The Hostel Utility Monitoring dashboard is a well-structured and informative tool that provides facility management with multi-dimensional visibility into energy consumption patterns. The analysis of the six dashboard visualizations reveals a facility with stable power quality, identifiable demand peaks, and a clear consumption hierarchy dominated by Sub_metering_3.

The data confirms that strategic interventions in lighting, HVAC scheduling, and off-peak load shifting can yield meaningful energy cost reductions. By acting on the recommendations in this

report — particularly addressing the 43% Sub_metering_3 share — the hostel can realistically target a 20–30% reduction in total electricity consumption within 12 months.

This dashboard should be maintained and expanded as the primary tool for ongoing utility management, with enhancements toward real-time monitoring and predictive analytics as the next evolution of the system.

— *End of Report* —

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